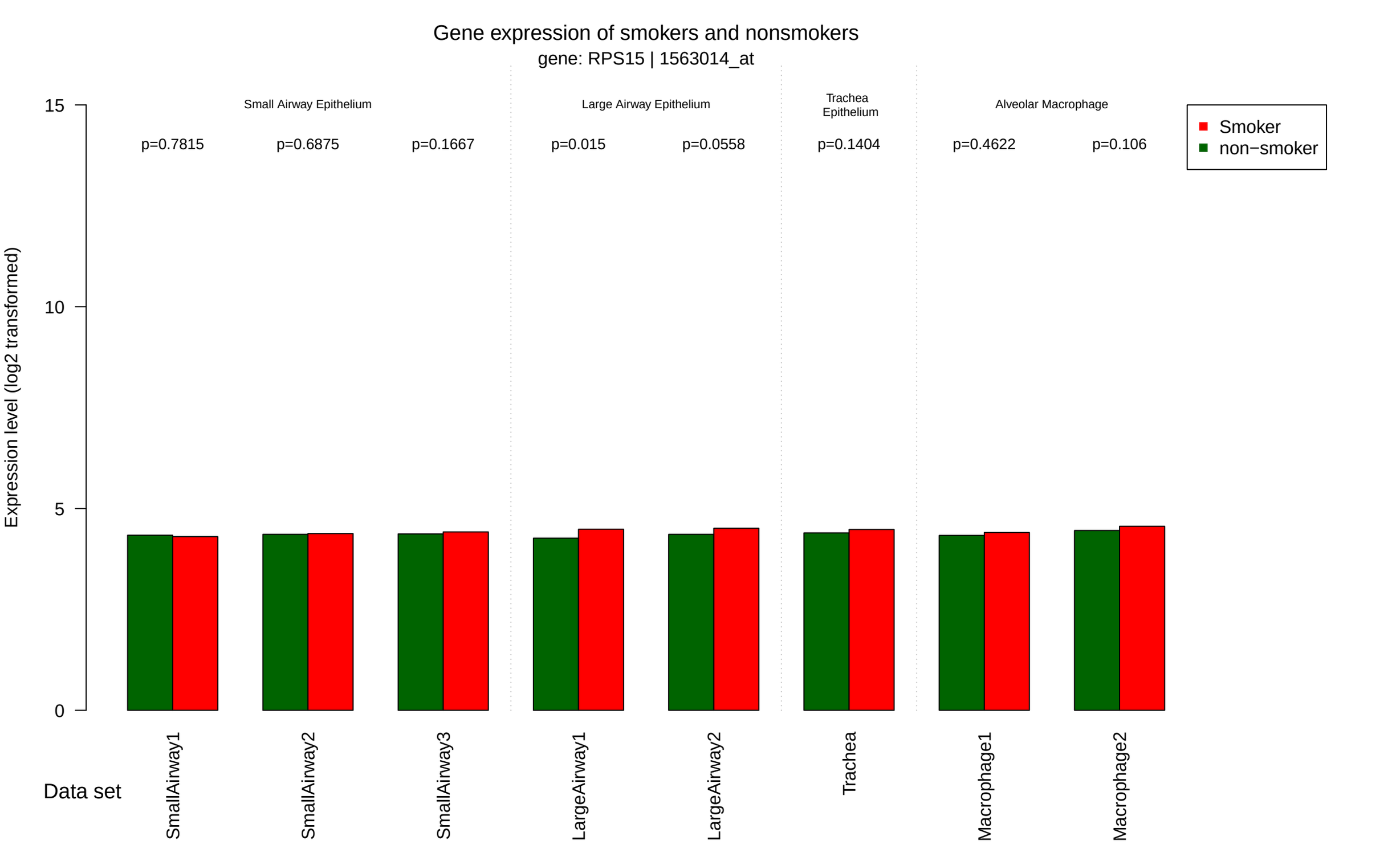
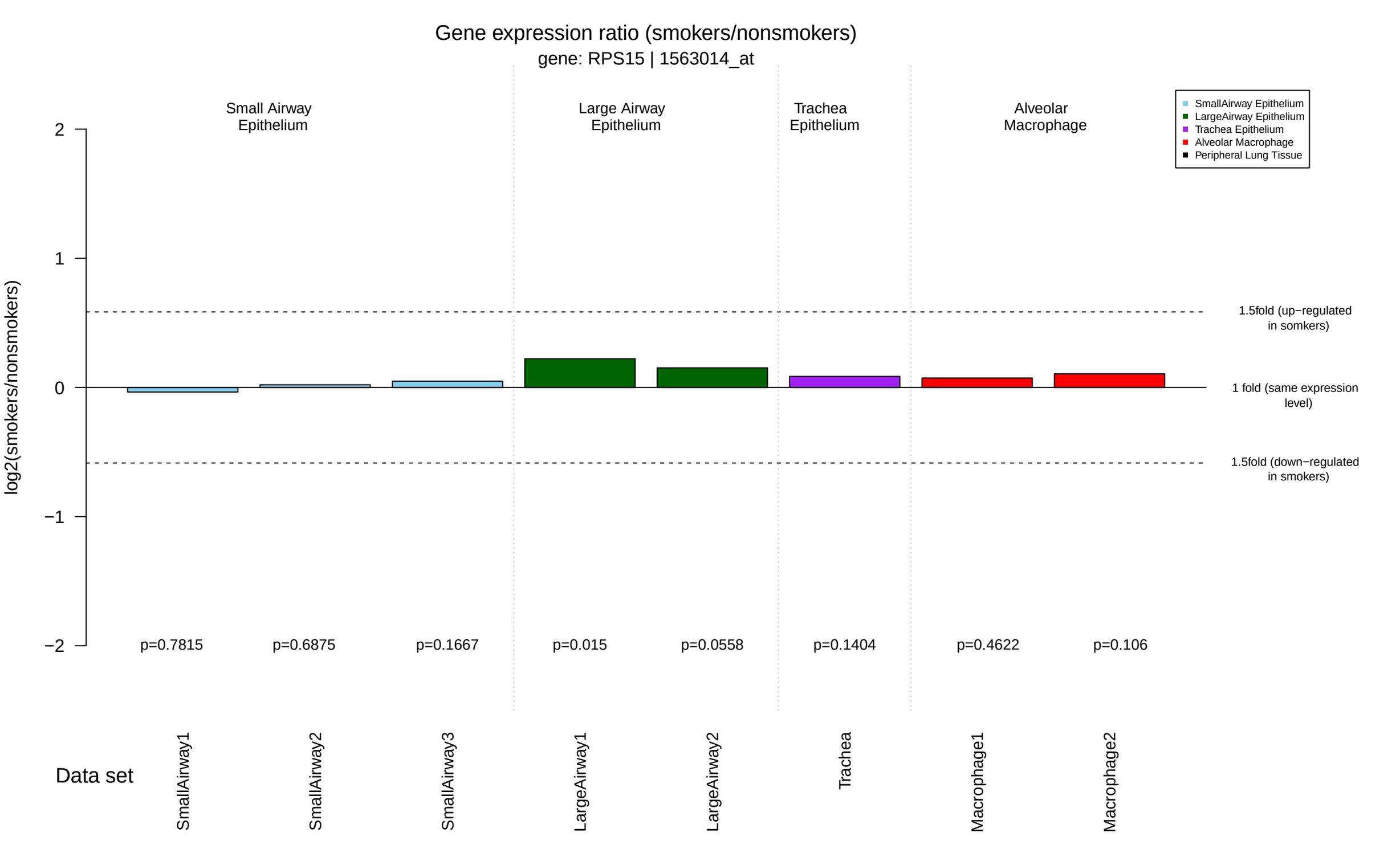
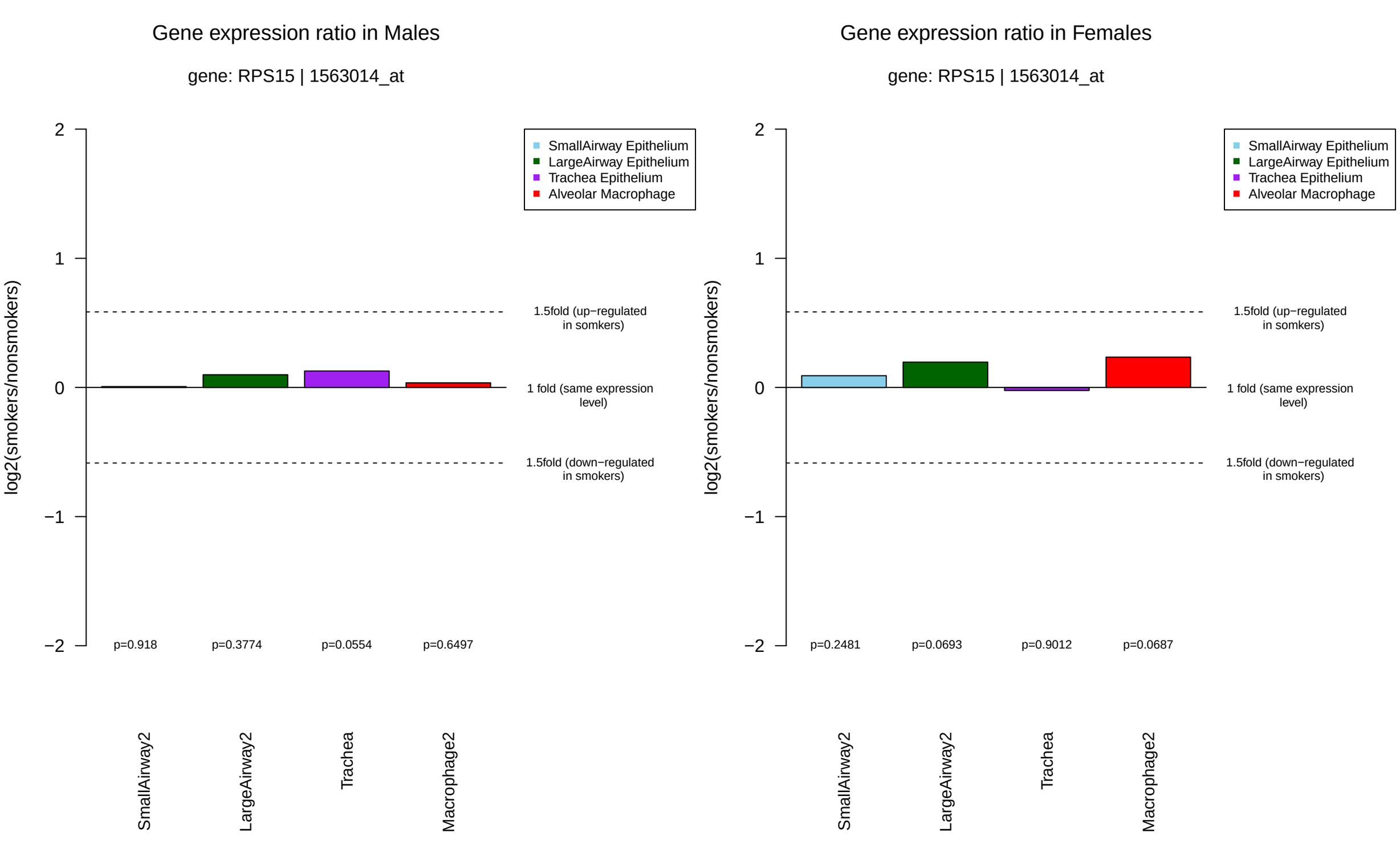
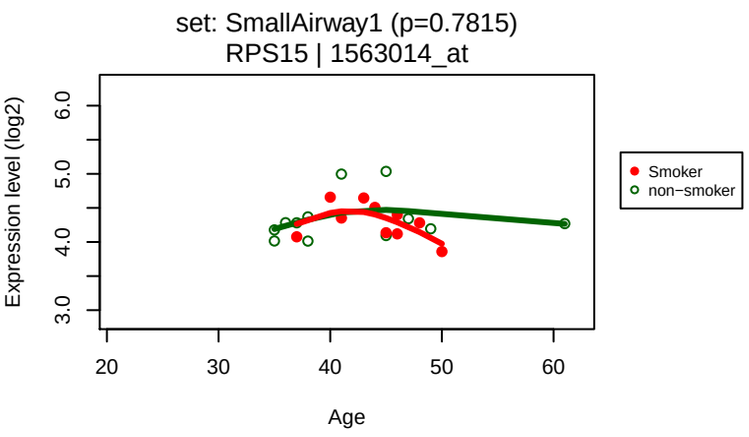
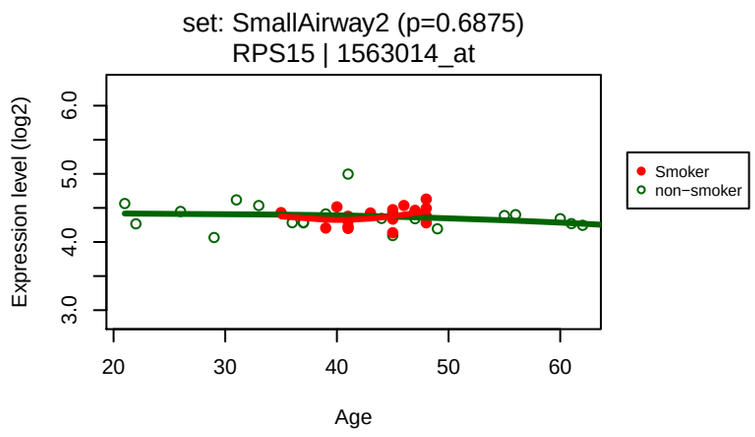

Gene expression of smokers and nonsmokers
gene: RPS15 | 1563014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7815
p=0.6875
p=0.1667
p=0.015
p=0.0558
p=0.1404
p=0.4622
p=0.106
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

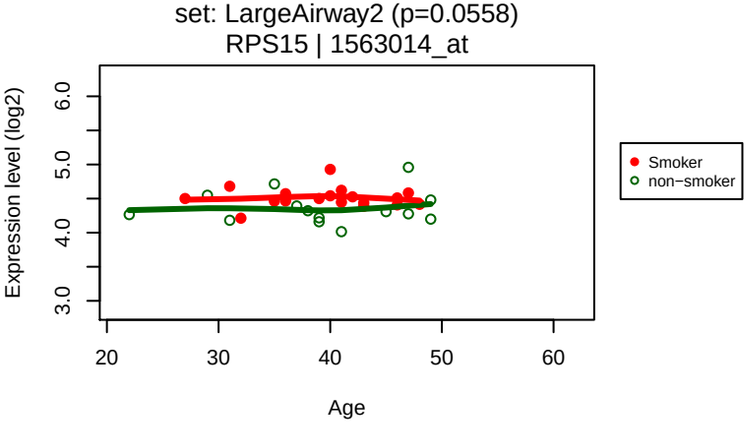

Gene expression ratio (smokers/nonsmokers)
gene: RPS15 | 1563014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7815
p=0.6875
p=0.1667
p=0.015
p=0.0558
p=0.1404
p=0.4622
p=0.106
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

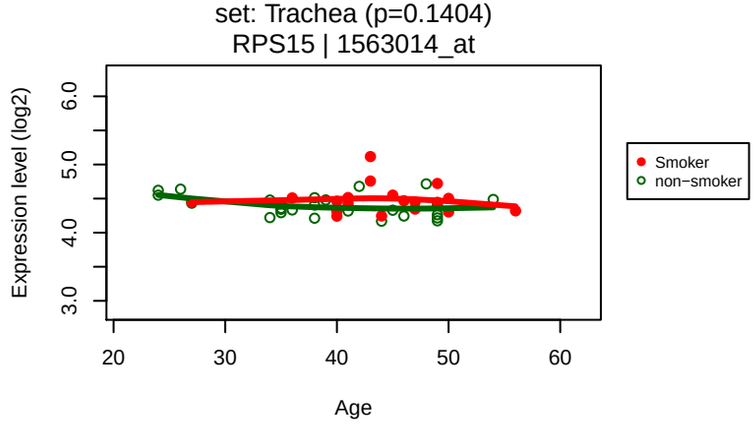

Gene expression ratio in Males
gene: RPS15 | 1563014_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.918
p=0.3774
p=0.0554
p=0.6497
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS15 | 1563014_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2481
p=0.0693
p=0.9012
p=0.0687
SmallAirway2
LargeAirway2
Trachea
Macrophage2

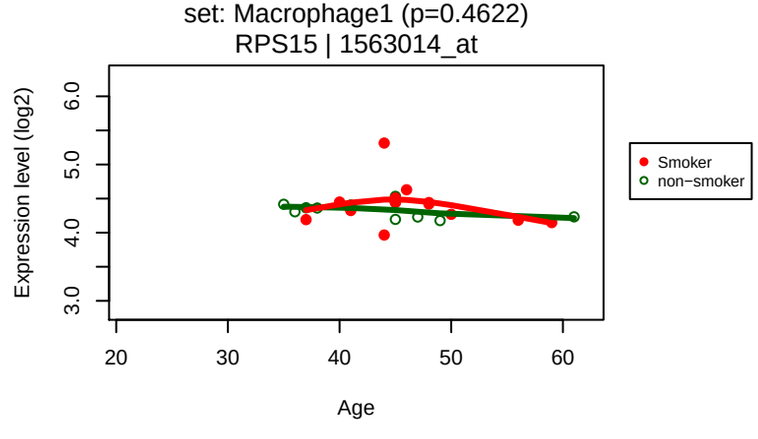

set: SmallAirway1 (p=0.7815)
RPS15 | 1563014_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker

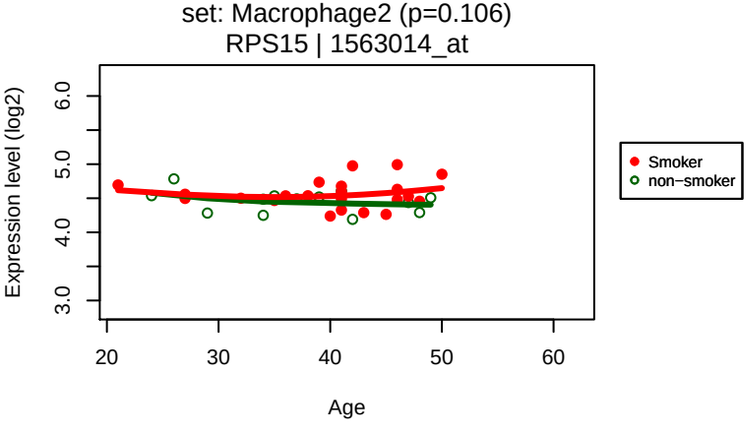

set: SmallAirway2 (p=0.6875)
RPS15 | 1563014_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker

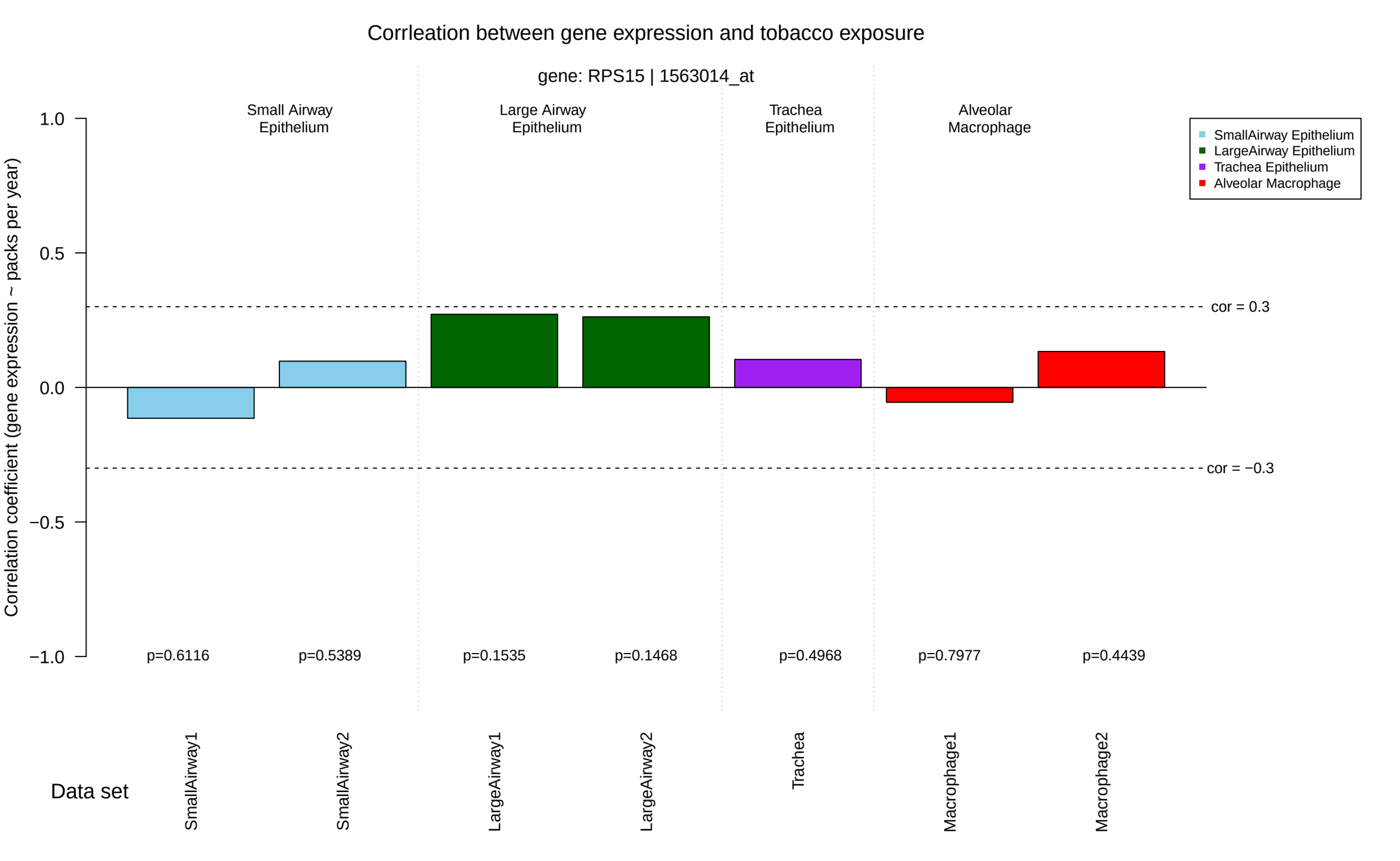
set: LargeAirway1 (p=0.015)
RPS15 | 1563014_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0558)
RPS15 | 1563014_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.1404)
RPS15 | 1563014_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4622)
RPS15 | 1563014_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.106)
RPS15 | 1563014_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS15 | 1563014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6116
p=0.5389
p=0.1535
p=0.1468
p=0.4968
p=0.7977
p=0.4439
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2